

software to implement optimization algorithms and solve problems computationally.

Text Books:

1. Chandra, S., Jyadeva, Mehra, A. Numerical Optimization. Narosa, 2009.
2. Bazaraa, M.S., Shefali, H.D., Shetty, C.M. Nonlinear Programming: Theory and Algorithms. New York: John Wiley & Sons, 1993. ISBN: 0471557935
3. Beck, A. Introduction to Nonlinear Optimization Theory, Algorithms, and Applications with MATLAB. SIAM Series on Optimization.

Reference Books:

1. Cottle, R.W., Thapa, M.N. Linear and Nonlinear Optimization. Springer, 2017.
2. Luenberger, D.G., Ye, Y. Linear and Nonlinear Programming, 5th Edition. Springer, 2021.
3. Bertsekas, D. Nonlinear Programming. Athena Scientific, 2016.
4. Boyd, S., Vandenberghe, L. Convex Optimization. Cambridge: Cambridge University Press, 2004.

Course Objectives: This course aims to provide the students with a basic understanding in the field of entrepreneurship, entrepreneurial perspectives, concepts and frameworks useful for analyzing entrepreneurial opportunities, understanding eco-system stakeholders and comprehending entrepreneurial decision making. It also intends to build competence with respect business model canvas and build understanding with respect to the domain of start-up venture finance.

Introduction to Entrepreneurship: Entrepreneurs; entrepreneurial personality and intentions - characteristics, traits and behavioral; entrepreneurial challenges.

Entrepreneurial Opportunities: Opportunities- discovery/ creation, Pattern identification and recognition for venture creation: prototype and exemplar model, reverse engineering.

Entrepreneurial Process and Decision Making: Entrepreneurial ecosystem, Ideation, development and exploitation of opportunities; Negotiation, decision making process and approaches, - Effectuation and Causation.

Crafting business models and Lean Start-ups: Introduction to business models; Creating value propositions - conventional industry logic, value innovation logic; customer focused innovation; building and analyzing business models; Business model canvas, Introduction to lean startups, Business Pitching.

Organizing Business and Entrepreneurial Finance: Forms of business organizations; organizational structures; Evolution of organization, sources and selection of venture finance options and its managerial implications. Policy Initiatives and focus; role of institutions in promoting entrepreneurship.

Course Learning Outcomes (CLOs) / Course Objectives (COs):

Upon successful completion of the course, the students should be able to:

1. Explain the fundamentals behind the entrepreneurial personality and their intentions
2. Discover/create and evaluate opportunities.
3. Identify various stakeholders for the idea and develop value proposition for the same.
4. Describe various Business Models and design a business model canvas.
5. Analyse and select suitable finance and revenue models for start-up venture.

Text Books:

Ries, Eric (2011), The lean Start-up: How constant innovation creates radically successful businesses, Penguin Books Limited.

Blank, Steve (2013), The Startup Owner's Manual: The Step by Step Guide for Building a Great Company, K&S Ranch.

S. Carter and D. Jones-Evans, Enterprise and small business- Principal Practice and Policy, Pearson Education (2006)

Reference Books:

1. T. H. Byers, R. C. Dorf, A. Nelson, Technology Ventures: From Idea to Enterprise, McGraw Hill (2013)
2. Osterwalder, Alex and Pigneur, Yves (2010) Business Model Generation.
3. Kachru, Upendra, India Land of a Billion Entrepreneurs, Pearson
4. Bagchi, Subroto, (2008), Go Kiss the World: Life Lessons For the Young Professional, Portfolio Penguin
5. Bagchi, Subroto, (2012). MBA At 16: A Teenager's Guide to Business, Penguin Books
6. Bansal, Rashmi, Stay Hungry Stay Foolish, CIIE, IIM Ahmedabad
7. Bansal, Rashmi, (2013). Follow Every Rainbow, Westland.
8. Mitra, Sramana (2008), Entrepreneur Journeys (Volume 1), Booksurge Publishing
9. Abrams, R. (2006). Six-week Start-up, Prentice-Hall of India.
10. Verstraete, T. and Laffitte, E.J. (2011). A Business Model of Entrepreneurship, Edward Elgar Publishing.
11. Johnson, Steven (2011). Where Good Ideas comes from, Penguin Books Limited.
12. Gabor, Michael E. (2013), Awakening the Entrepreneur Within, Primento.
13. Guillebeau, Chris (2012), The \$100 startup: Fire your Boss, Do what you love and work better to live more, Pan Macmillan
14. Kelley, Tom (2011), The ten faces of innovation, Currency Doubleday
15. Prasad, Rohit (2013), Start-up sutra: what the angels won't tell you about business and life, Hachette India.